

1) The equivalent resistance R_{eq} between the terminals a-b of the current source is:

- a) 6Ω b) 6.67Ω
c) 18Ω d) 12Ω

2) The current I_o in the 6Ω resistor is:

- a) $4A$ b) $2A$
c) $10A$ d) $6A$

3) The voltage across 8Ω resistor is:

- a) $120V$ b) $60V$
c) $150V$ d) $240V$

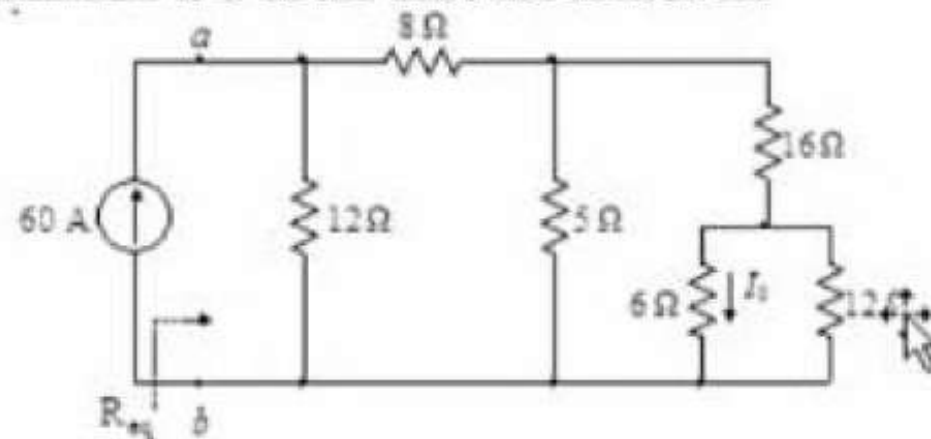


Fig.1

4) The value of V_1 is:

- a) 36.5V
- b) 43.75V
- c) 365 mV
- d) -49.8V

5) The value of V_2 is:

- a) 5V
- b) 200mV
- c) 25V
- d) 50V

6) The value of the current (i) in $2\ \Omega$ resistor is:

- a) 6.45A
- b) 4.48A
- c) -4.15A
- d) 8.75A

7) The power absorbed by $4\ \Omega$ resistor is:

- a) 156.25W
- b) 10.46W
- c) 50.23W
- d) 64.46W

8) The power delivered by 5 A current source is:

- a) 4W
- b) 12.5W
- c) 25W
- d) -44.8W

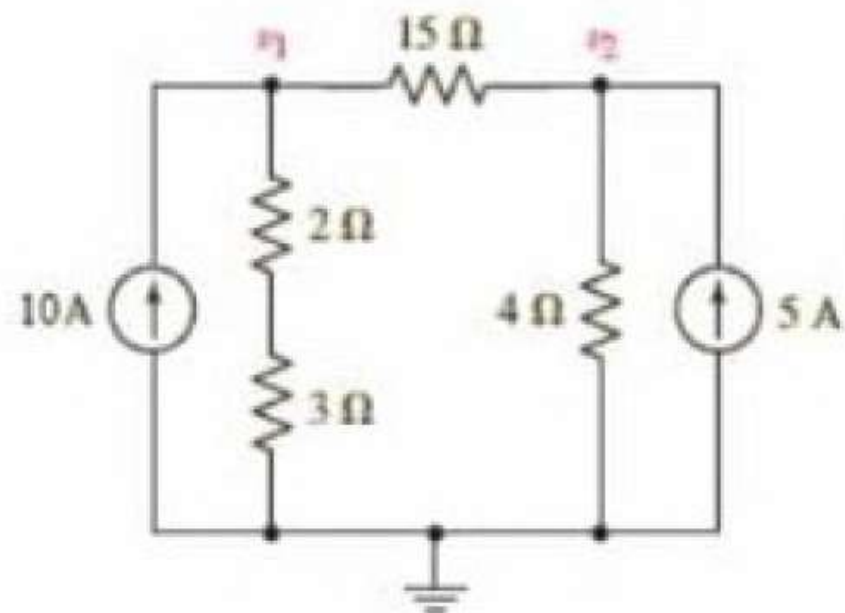


Fig.2

9) The value of V_x is:

- a) 25V **b) 32V**
c) 64V d) 50V

10) The current in $20\ \Omega$ resistor is:

- a) 1.9A b) 128 mA
c) 1.6 A d) 2.8A

11) The delivered power by the 8 A current source is: _____

- a) 60W b) 160 W
c) 120 W **d) 256W**

12) Refer to figure 4; the value of the equivalent resistance is:

- a) 33.6 Ω **b) 44.375 Ω**
c) 56.5 Ω d) 75 Ω

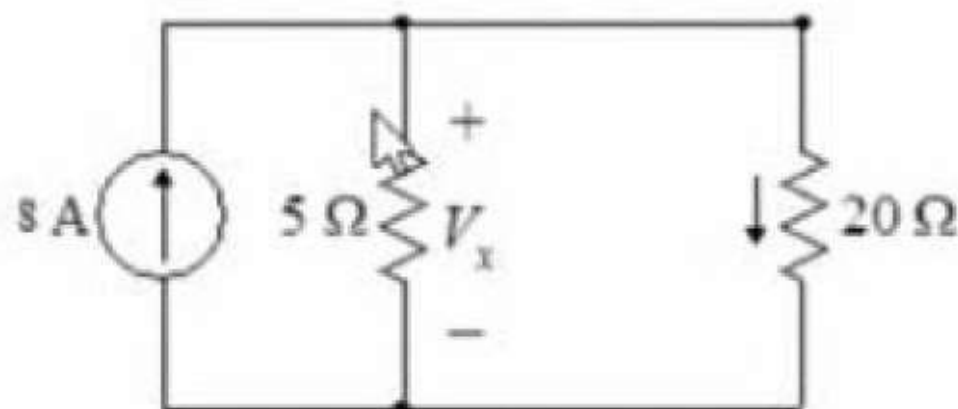


Fig.3

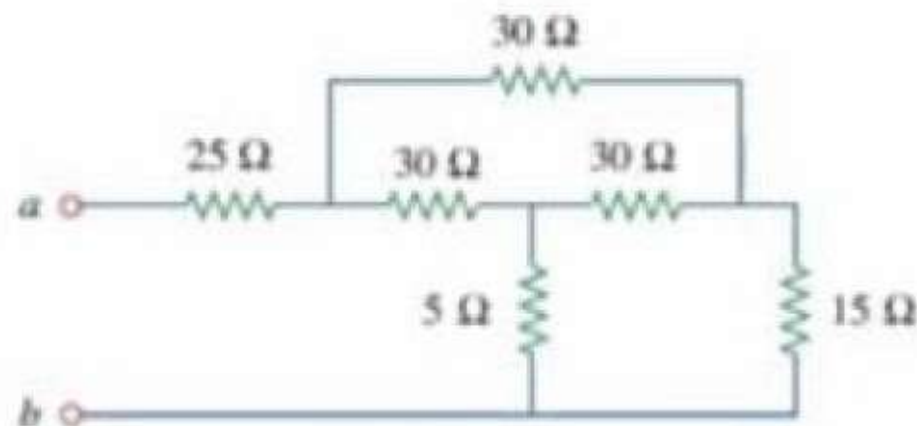


Fig 4

13) How many joules of energy will a $10\ \Omega$ resistor connected across a 15V battery dissipate in 1 minute?

- a) 13.5kJ b) 1.35kJ c) 0.135kJ d) 135 J

14) A charge of 2 C flowing past a given point each milli second is a current of:

- A) 2M A b) 200 mA
c) 2 mA d) 2 kA

15) How many coulombs of charge flow through a circuit carrying a current of 10 A in 1 minute:

- a) 10 C b) 160 C
c) 600 C d) 1200 C

Q1: Find R_{eq} , I_o , V_x

$$R_{(1)} = \frac{6 \times 12}{6 + 12} = 4 \Omega$$

$$R_{(2)} = 4 + 16 = 20 \Omega$$

$$R_{(3)} = \frac{20 \times 5}{20 + 5} = 4 \Omega$$

$$R_{(4)} = 4 + 8 = 12 \Omega$$

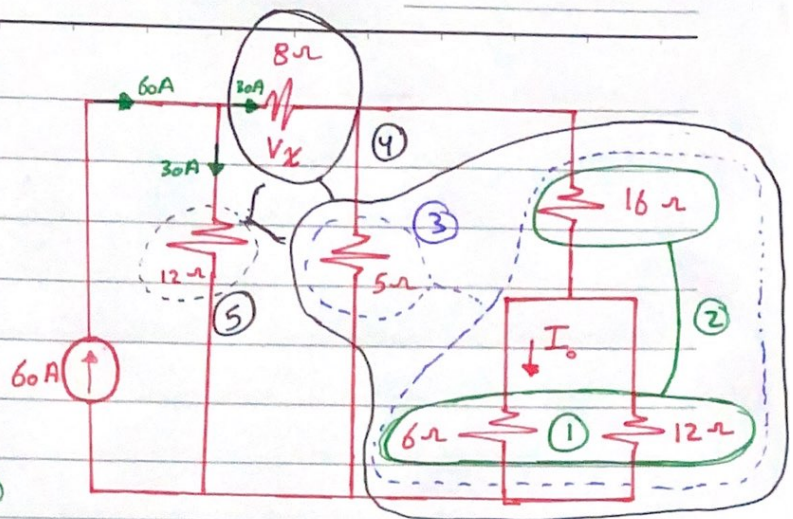
$$R_{(5)} = \frac{12 \times 12}{12 + 12} = 6 \Omega$$

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$$R = 12, V = 360, I = 30$$

$$R = 8, V = 240, I = 30$$

$$R_{(3)} = 4, I = 30, V = 120$$



$$R_{eq} = 6 \Omega$$

$$I_s = 60 A$$

$$V = 60 \times 6 = 360 V$$

$$R_{(2)} = 20, V = 120, I = 6$$

$$R = 16, I = 6, V = 96$$

$$R_{(1)} = 4, I = 6, V = 24$$

$$R = 6, I = 4, V = 24$$

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Q<sub>2</sub> ∴ Find Value  $V_1, V_2, I_2, P_{4\Omega}, P_{5A}$

at node ① :

$$10 = I_1 + I_2$$

$$10 = \frac{V_1 - V_2}{15} + \frac{V_1}{5} \quad (*15)$$

$$150 = V_1 - V_2 + 3V_1$$

$$150 = 4V_1 - V_2 \Rightarrow V_1 = \frac{150 + V_2}{4}$$

at node ② :

$$I_1 + 5 = I_3$$

$$\frac{V_1 - V_2}{15} + 5 = \frac{V_2}{4}$$

$$\frac{V_1 - V_2}{15} - \frac{V_2}{4} = -5$$

$$\frac{4V_1 - 4V_2 - 15V_2}{60} = -5$$

$$4V_1 - 19V_2 = -300$$

$$4\left(\frac{150 + V_2}{4}\right) - 19V_2 = -300$$

$$150 - 18V_2 = -300$$

$$450 = 18V_2$$

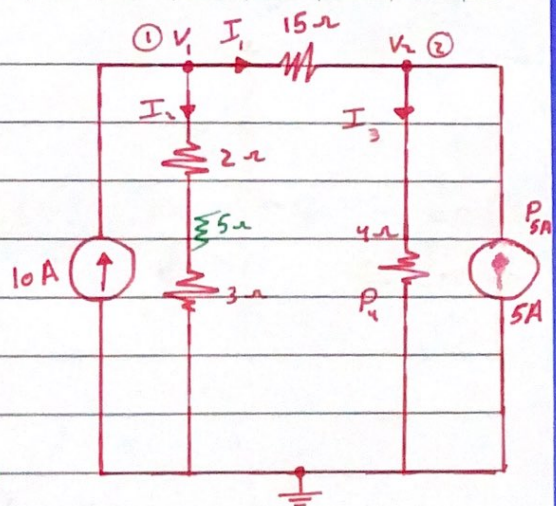
$$① \quad 25 = V_2$$

$$② \quad V_1 = \frac{175}{4} = 43.75$$

$$I_2 = \frac{V_1}{5} = \frac{43.75}{5} = 8.75 \text{ A} \quad ③$$

$$P_4 = I_3 V = (I_2)^2 R = \left(\frac{25}{4}\right)^2 (4) = 156.25 \text{ W} \quad ④$$

$$P_{5A} = I_5 V = (5)(25) = 125 \text{ W} \quad ⑤$$



ناتج الـ 5A  
ناتج الـ 4Ω

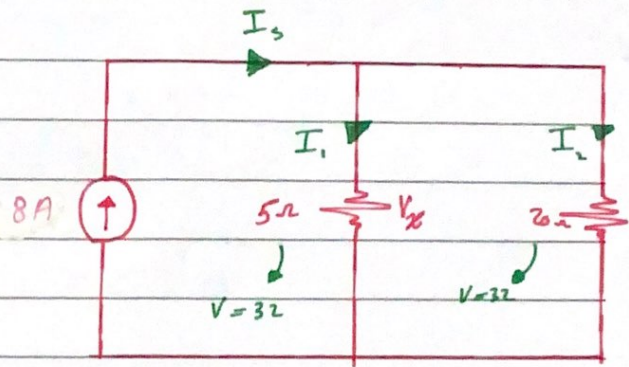
Q<sub>3</sub>: Find  $V_x$ ,  $I_2$ ,  $P_{8A}$

$$R_{eq} = \frac{5 \times 20}{5 + 20} = 4 \Omega$$

$$V = (8)(4) = \boxed{32 \text{ V}} \quad (1) \checkmark$$

$$I_2 = \frac{V}{R_2} = \frac{32}{20} = \boxed{1.6 \text{ A}} \quad (2) \checkmark$$

$$P_{8A} = (32)(8) = \boxed{256} \quad (3) \checkmark$$



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Q<sub>4</sub>: How many Joules of energy will a (10 Ω) resistor connected across a (15V) Battery dissipate in (1) minutes?

$$W = Pt \quad \text{But: } p = iv \Rightarrow p = \frac{v^2}{R}$$

$$W = \left( \frac{v^2}{R} \right) t$$

$$W = \frac{(15)^2}{10} (60) = 1350 \text{ KJ}$$

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Q<sub>5</sub>: A charge of (2C) flowing past a given point each (milli second) is a current of:

$$Q = \underline{2C}, \quad t = 1 \times 10^{-3} \text{ s}, \quad i = \frac{q}{t}$$

$$i = \frac{2}{1 \times 10^{-3}} = 2 \times 10^3 \text{ A}$$

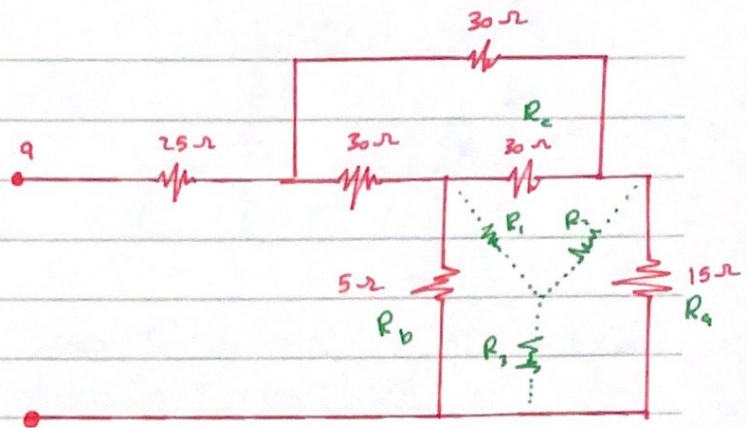
Q<sub>6</sub>: How many coulombs of charge flow through a circuit carrying a current of (10) A in (1) minutes?

$$i = 10 \text{ A}, \quad t = 60, \quad Q = it$$

$$Q = (10)(60) = 600 \text{ C}$$

Q7: Find The  $R_{eq}$

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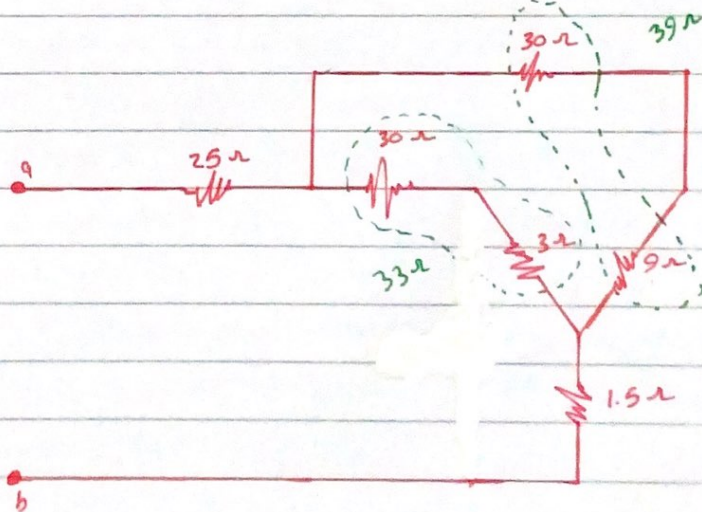


$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{5 * 30}{15 + 5 + 30} = 3 \Omega$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} = \frac{15 * 30}{50} = 9 \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 * 5}{50} = 1.5 \Omega$$

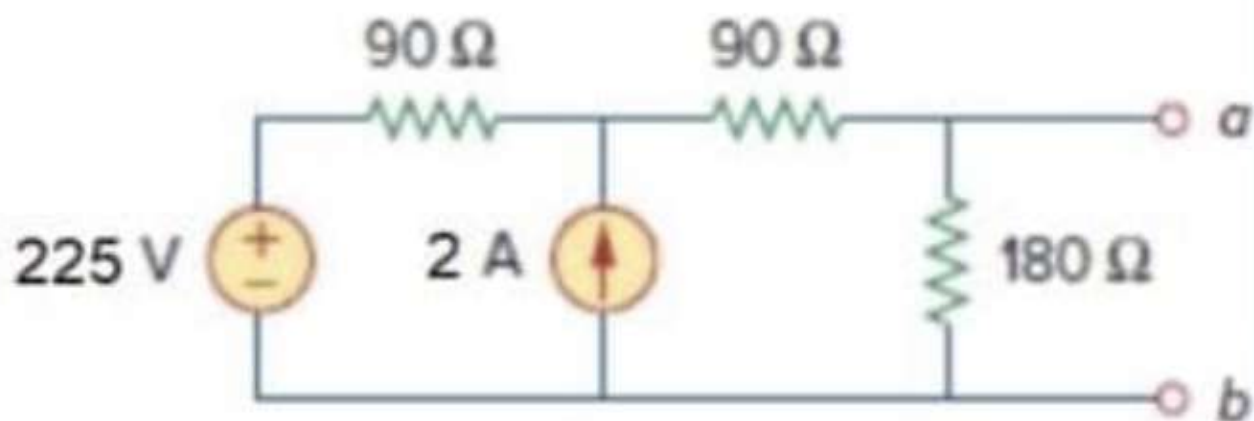
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$$\frac{33 * 39}{33 + 39} = \frac{143}{8}$$

$$\frac{143}{8} + 1.5 + 25 = 44.375$$

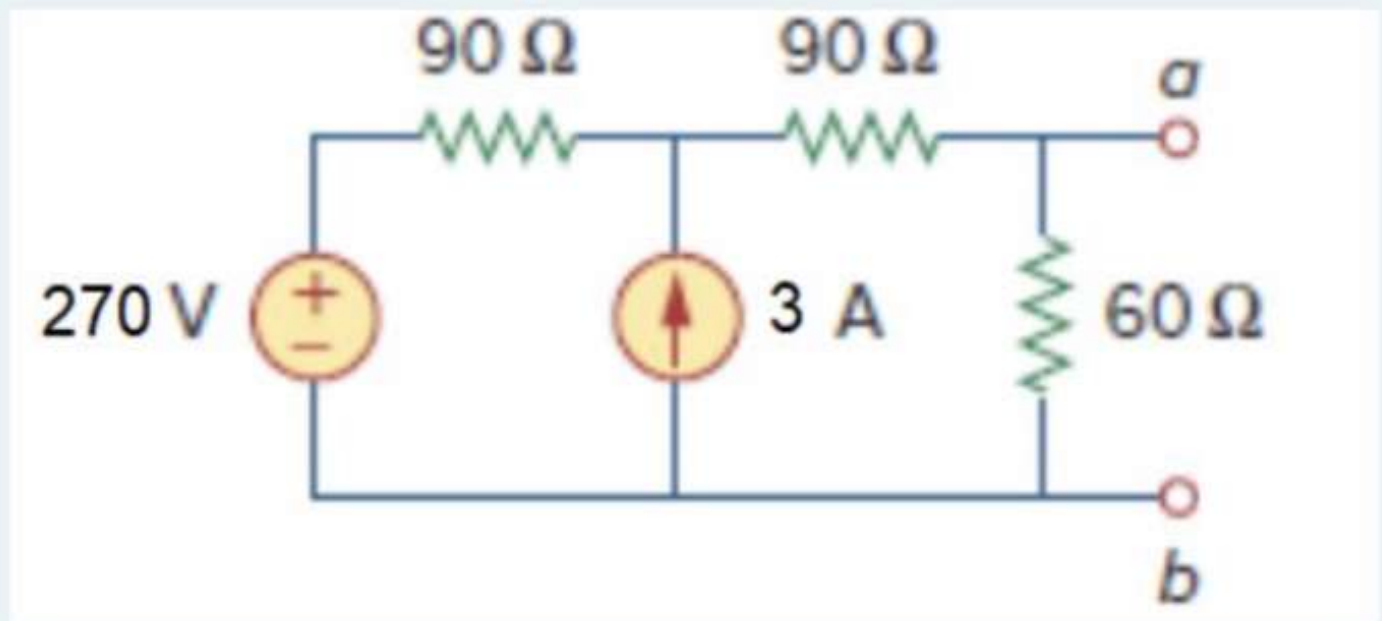
Find the Norton equivalent circuit to the left of terminals in the circuit of the figure.



- ☐ a. None of all
- ☐ b. 2.25A, 90Ω
- ☒ c. 1.125A, 90 Ω
- ☐ d. 1.125 A, 180 Ω
- ☐ e. 6.75A, 45 Ω

*Laith Q Amro* ↵

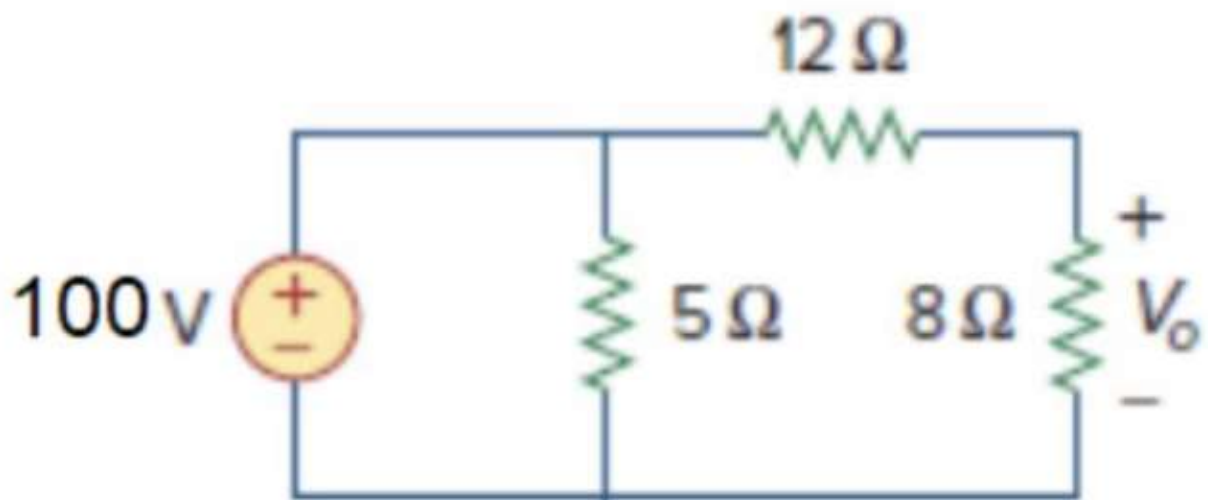
Find the Thevenin equivalent circuit to the left of terminals in the circuit of the figure.



- ☒ a. 135V, 45Ω
- ☐ b. 64 V, 180 Ω
- ☐ c. 160V, 90 Ω
- ☐ d. 140 V , 45 Ω
- ☐ e. None of all

*Laith Q Amro* ◀

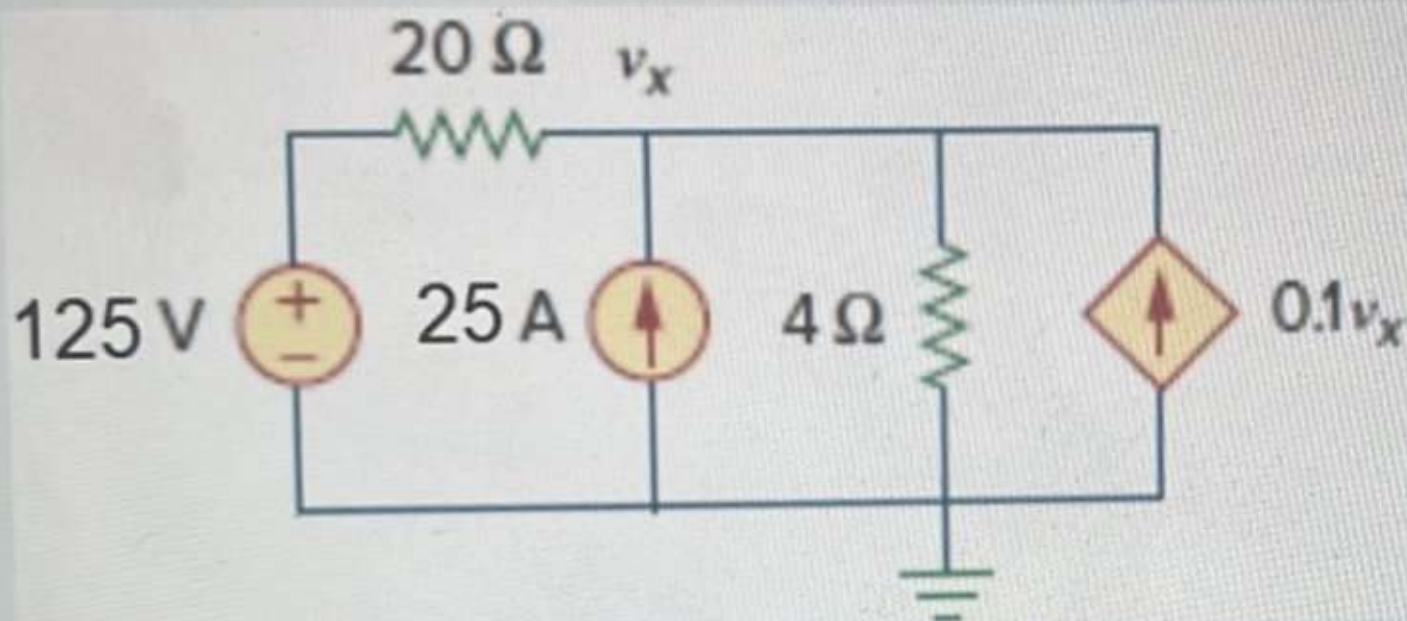
Refer to the figure and assume  $V_o=1V$  and use linearity to calculate the actual value of  $V_o$ .



- ☒ a. 40 V
- ☐ b. 64 V
- ☐ c. 16 V
- ☐ d. None of all
- ☐ e. 32 V

*Laith Q Amro*

Using superposition theorem For the the circuit shown in figure.



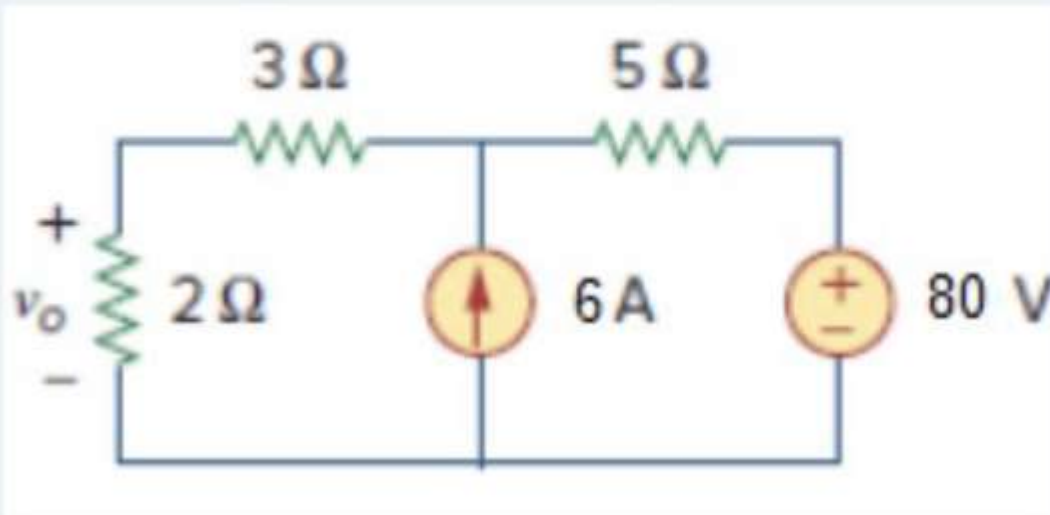
The value of  $V_x$  is :

- ☐ a. 50 V
- ☐ b. 125 V
- ☒ c. 156.25 V
- ☐ d. 64.5 V
- ☐ e. None of all

Clear my choice

Laith Q Amro |

Using superposition theorem For the the circuit shown in figure.



The value of  $V_o$  is :

☐ a.  $11.45\ \text{V}$

☒ b.  $22\ \text{V}$

☐ c.  $-4.15\ \text{V}$

☐ d.  $4.48\ \text{V}$

☐ e. None of all

*Laith Q Amro*